Elias BENGUEDDA LAUZON

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Quantitative Research / Trading Intern – 6-month internship (gap year) starting January 2027

Professional Experience

Société Générale – Pricing Team, Paris **Jun 2026 – Dec 2026**
Quantitative Research Intern – Derivatives Pricing

- Implemented a **callable bond** pricer under **Hull–White 1F**: Jamshidian decomposition for European calls, **Monte Carlo + Longstaff–Schwartz** for Bermudan and American calls
- Built and backtested correlation models (**EWMA, DCC-GARCH, Ornstein–Uhlenbeck**) for pricing **Worst-of / Best-of** basket options; **EWMA model adopted by the desk**

Natixis – Trading Solutions, Paris **Summer 2025**
Quantitative Analyst Intern – Equity & FICC

- Automated **P&L** monitoring with Python data pipelines, moving the follow-up from **quarterly to daily**
- Built position tracking and performance analysis tools on **equity derivatives** for front-office traders

Quantitative Projects

Research papers & code: **portfolio-eliasbenguedda.fr**

Heston & Local Stochastic Volatility Calibration – *CAC40 options, Python*

- Calibrated **Heston** to 142 CAC40 options via Fourier pricing: implied vol RMSE 0.42, down to 0.12 with time-dependent parameters
- Calibrated an **LSV** model (Dupire local vol, particle method) and quantified **model risk** on barriers and autocalls at identical vanilla prices

Autocallable Notes Pricer – *Athena, Phoenix, Monte Carlo*

- Built a **Monte Carlo** pricer under calibrated Heston with variance reduction, validated on 5 closed-form limits
- Computed fair coupons, call/loss probabilities and **Greeks** (vega, dividends) for 35 product variants

Interest Rate Curves & Derivatives – *€STR, EURIBOR, SOFR, Hull–White*

- Bootstrapped €STR OIS and dual-curve EURIBOR **yield curves** (QuantLib-matched); PCA on 181 Treasury curves
- Priced a **callable SOFR range accrual** under Hull–White (static replication, Monte Carlo + **Longstaff–Schwartz**)

Education

Mines Saint-Étienne (Grande École d'ingénieurs), France **2024 – 2028**
MSc in Engineering (Diplôme d'Ingénieur), Applied Mathematics & Data Science gap year 2026–27
Probability, stochastic processes, optimization, machine learning, statistical modelling

École Normale Supérieure de Lyon (ENS de Lyon), France **2025 – 2026**
Advanced courses in Machine Learning & Stochastic Modelling
Reinforcement learning, probabilistic modelling, convex optimization, statistical learning theory

Aix-Marseille University, France **2025 – 2026**
Master's in Fundamental Mathematics
Stochastic calculus, PDEs, numerical methods, measure theory

Lycée Janson de Sailly, Paris **2022 – 2024**
Classes Préparatoires MPSI/MP – intensive Mathematics, Physics & Computer Science
Ranked 1st in mathematics (2nd year); admissible to the Centrale and Mines entrance exams

Skills, Languages & Interests

- **Programming:** Python (NumPy, SciPy, Pandas, scikit-learn, PyTorch), SQL, Git, LaTeX; C++ (basic), C# (basic)
- **Tools:** QuantLib, Numerix, Bloomberg Terminal
- **Languages:** French (native), English (fluent), Spanish (basic)
- **Interests:** Karting (17th in the French National Championship), Aviation (student pilot)